



File Handling in C++

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Programming Fundamentals



Filing in C++

- "file handling" or "file I/O" in the context of programming, it refers to the process of reading from and writing to files on a computer's storage system.
- In C++ and many other programming languages, file handling is typically achieved through libraries that provide classes and functions for working with files.
- The `<fstream>` library in C++ is commonly used for this purpose.

Create and Write To a File:

- To create a file, use the ofstream or class, and specify the name of the file.
- To write to the file, use the insertion operator (<<).

```
elloWorld.cpp
#include <iostream>
#include <fstream>

using namespace std;
int main(){

    // Create and open a text file
    ofstream MyFile("first_file.txt");
    // Write to the file
    MyFile << "Wrting hello word in text file";
    // Close the file
    MyFile.close();

    return 0;
}
```

Key File Handling Classes:



ifstream

Input File Stream. Used specifically for reading data from a file. Think of it as ``cin`` for files.



ofstream

Output File Stream. Used specifically for writing data to a file. Think of it as ``cout`` for files.



fstream

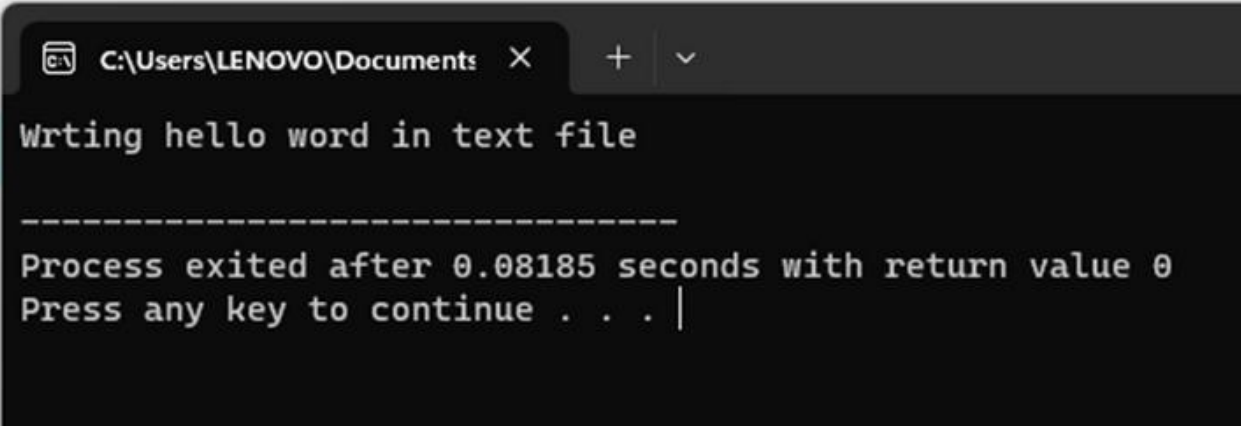
File Stream. The general-purpose class. It can be used for both reading from and writing to files.

Read a File:

- To read from a file, use the `ifstream` class, and the name of the file.
- Note that we also use a while loop together with the `getline()` function (which belongs to the `ifstream` class) to read the file line by line, and to print the content of the file:

Example:

```
helloWorld.cpp  read.cpp
1  #include <iostream>
2  #include <fstream>
3
4  using namespace std;
5  int main(){
6
7      string myText;
8      // Read from the text file
9      ifstream MyReadFile("first_file.txt");
10     // Use a while loop together with the getline() function
11     while (getline (MyReadFile, myText)) {
12         // Output the text from the file
13         cout << myText <<endl;
14     }
15     // Close the file
16     MyReadFile.close();
17
18     return 0;
19 }
```



Reading a File Line by Line:

The ``getline`` Function

- The `>>` operator stops at spaces. To read an entire line of text, use the `getline()` function.
- This is commonly combined with a while loop to read an entire file, line by line, until the end.



End Of Class

